

Michael Mulugeta Demissie

Data Engineer | ML Infrastructure · ETL Pipelines · Cloud · Python

Washington, DC | michaelmulugetademissie@gmail.com | (202) 290-6215 | LinkedIn | GitHub | Portfolio

SUMMARY

Data Engineer with hands-on experience building production ETL pipelines, deploying ML models, and shipping full stack data applications. Skilled across the full data lifecycle — from ingestion and transformation to model serving and API delivery.

EDUCATION

The George Washington University — M.S. Data Analytics

Expected May 2027

Delhi Technological University — B.Tech Software Engineering

May 2025

TECHNICAL SKILLS

Data Engineering: Python, SQL, Airflow, Spark, Kafka, dbt, ETL/ELT, Data Modeling

Cloud & Infra: AWS (S3, Glue, Redshift, Lambda), Snowflake, BigQuery, Terraform, Docker

Databases: PostgreSQL, MySQL, Redis, Elasticsearch, MariaDB

ML & AI: PyTorch, TensorFlow, scikit-learn, HuggingFace, BERT, OpenAI API, MLflow

Full Stack: FastAPI, Flask, Node.js, Express, React, JavaScript, Bash, Git, CI/CD, JWT Auth

Visualization: Tableau, Power BI, Plotly, Dash, Grafana, Looker

EXPERIENCE

Data Engineer — Hasab AI

Oct 2025 – Apr 2026

Remote

- Architected Python ETL pipelines processing multilingual audio datasets across 10+ African and Middle Eastern languages, handling segmentation, noise filtering, and feature extraction.
- Built and deployed FastAPI services exposing structured NLP data to downstream speech and voice AI applications in underserved language markets.

Data Engineer — ParallelDots

Sep 2024 – Jan 2025

Delhi, India

- Engineered Python ETL pipelines ingesting retail shelf image metadata into MySQL, powering the ShelfWatch AI CV platform used by Fortune 500 FMCG clients globally.
- Built Pandas/NumPy preprocessing pipelines for CV model training and developed Flask REST APIs exposing processed data to production dashboards.

TECHNICAL PROJECTS

FinResolve — Complaints Data Pipeline & ML Routing System

Jan 2026 – May 2026

Python · Airflow · PostgreSQL · AWS S3 · Docker · FastAPI · DistilBERT · Render

- Built end-to-end ETL pipeline processing 1M+ CFPB records; engineered 812K training samples across 9 departments for ML routing.
- Fine-tuned DistilBERT for department classification and trained TF-IDF priority classifier achieving 88% accuracy; deployed via Docker on Render with JWT-auth FastAPI and role-based dashboards.

DC Metro Real-Time Arrivals Data Pipeline

May 2026

Node.js · PostgreSQL · Airflow · Docker · Express · Railway · WMATA API

- Built real-time ingestion pipeline polling WMATA API every 60s across all DC Metro stations, sustaining 580K+ rows/day with Airflow-scheduled rolling 24hr retention.
- Designed full architecture end-to-end: poller → PostgreSQL → Express API → live dashboard; containerized with Docker, deployed to production on Railway.